

KCPC Week 1 Solutions

1 Logic

Basic – Same Birthday Month

There are only 12 months. If 13 students are placed into 12 month-boxes by birth month, the pigeonhole principle forces at least one box to contain at least two students.

Easy – Glove Selection

For one matching pair, the adversarial draw is all gloves of the same hand: $5 + 3 + 2 = 10$ gloves. The next glove guarantees a pair, so the answer is 11.

For one matching pair of each colour, the worst case is all 10 black gloves, all 6 brown gloves, and 2 grey gloves of the same hand. The next grey glove completes the final pair, so the answer is 19.

Medium – Three People: Knights, Knaves, and Spies

Let the types be K (Knight), N (Knave), S (Spy). Exactly one of each exists. Alice says “Bob is N ”; Bob says “Carol is K ”; Carol says “Alice is S ”.

Suppose Alice is K . Then Bob is N , so Bob lies. Bob says “Carol is K ”, so Carol is not K . Carol must be S . Then Alice is K , Bob is N , Carol is S – consistent. Carol says “Alice is S ”, which is false; as Spy she may lie. This works.

Suppose Bob is K . Then Carol is K , contradiction (two Knights).

Suppose Carol is K . Then Alice is S , so Alice is not K . Bob must be K (only one K), but then Carol is K – contradiction.

The only consistent assignment: Alice = Knight, Bob = Knave, Carol = Spy.

Key insight: Test each possible assignment of the unique Knight; contradictions narrow it down.

Hard – Islander Statements

Let L be the number of liars. Statement i is true exactly when $L \geq i$. Therefore the true statements are precisely $1, 2, \dots, L$, so there are L truth-tellers and $10 - L$ liars. But L is the number of liars, so $L = 10 - L$, hence $L = 5$. Islanders 1 through 5 tell the truth; islanders 6 through 10 lie.

Very Hard – Counterfeit Coin

One standard strategy is as follows. Label the coins 1 to 12. First weigh 1, 2, 3, 4 against 5, 6, 7, 8.

If they balance, the counterfeit is among 9, 10, 11, 12. Weigh 9, 10, 11 against 1, 2, 3. If this balances, coin 12 is counterfeit; compare it with coin 1 to decide heavy/light. If it does not balance, the odd coin is one of 9, 10, 11 and its direction is known; weigh 9 against 10 to identify it.

If the first weighing is unbalanced, suppose 1, 2, 3, 4 are heavier than 5, 6, 7, 8; the other case is symmetric. Then the counterfeit is either heavy among 1, 2, 3, 4 or light among 5, 6, 7, 8. Weigh 1, 2, 5 against 3, 6, 9. If left is heavy, the counterfeit is heavy 1 or 2, or light 6; weigh 1 against 2 to decide. If right is heavy, the counterfeit is heavy 3 or light 5; weigh 3 against 9 to decide. If balanced, the counterfeit is heavy 4, light 7, or light 8; weigh 7 against 8 to decide.

2 Invariants

Basic – Jigsaw Assembly

Each move reduces the number of sections by exactly 1. Starting with 500 sections and ending with 1 requires 499 moves.

Easy – Odd Handshakes

Sum the number of handshakes made by each person. This total equals twice the number of handshakes, so it is even. The people with even handshake counts contribute an even amount to the sum. Therefore the people with odd handshake counts must also contribute an even amount. A sum of odd numbers is even only when there are an even number of them.

Medium – Chameleons

Track the differences between colour counts modulo 3. A meeting changes two colour counts by -1 and the third by $+2$, so differences are unchanged modulo 3. Initially the differences include $14 - 10 = 4 \equiv 1$, $15 - 14 = 1 \equiv 1$, and $15 - 10 = 5 \equiv 2 \pmod{3}$. If all colours became equal except one colour present, some relevant difference would be $0 \pmod{3}$. The invariant prevents this, so it is impossible.

Hard – The MU Puzzle

Track the number of I's modulo 3. Initially it is 1. Rules 1 and 4 do not change it, Rule 2 doubles it, and Rule 3 subtracts 3. Starting from a number not divisible by 3, these operations never make it divisible by 3. The string MU has 0 I's, divisible by 3, so it is unreachable.

Very Hard – The Fifteen Puzzle

Associate a board with the permutation of tiles read row by row. Legal horizontal moves preserve inversion parity; legal vertical moves change inversion parity and also change the blank row parity. The combined invariant is preserved. Swapping 14 and 15 changes inversion parity but leaves the blank position unchanged, so the target is unreachable.

3 Colouring Arguments

Basic – One Missing Corner

No. A domino covers two squares, so any domino tiling covers an even number of squares. The board has 63 squares.

Easy – Corner-To-Corner Knight Journey

No. A knight alternates square colour every move. Visiting all 64 squares uses 63 moves, an odd number, so the start and end squares would need opposite colours. Opposite corners of a chessboard have the same colour.

Medium – Board Walks

Colour the shapes like a chessboard. A path using horizontal/vertical moves alternates colours. If the counts differ by more than 1, no Hamilton path exists. For the first cross, the colour classes have 21 and 24 squares, so no path exists.

For the second cross, the following diagram gives a path: visit the numbered squares in order from 01 to 41. Blanks lie outside the cross; -- marks the four removed squares. Consecutive numbers are horizontally or vertically adjacent, and every remaining square is used.

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      -- 35 36
      33 34 37
      32 39 38
-- 05 06 31 40 41 26 25 --
01 04 07 30 29 28 27 24 23
02 03 08 09 18 19 20 21 22
      10 17 16
      11 12 15
      -- 13 14

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Hard – Jumping To The Other Side

Colour the board as a chessboard. Every legal jump lands on a square of the same colour as the starting square: in each allowed direction, it travels two steps and the jumped counter stays put. Initially there are 9 counters on one colour among the A cells, but only 6 B cells of that colour. The number of counters of each colour is invariant, so the task is impossible.

Very Hard – Domino Tiling Revisited

The answer is: exactly even n . Odd n leaves an odd number of remaining squares after removing two squares, so tiling is impossible. For even n , trace a cycle through every square of the board: cross the top row from left to right, snake back and forth through the remaining rows using columns 2 through n , then return up column 1 to the starting square. Consecutive squares on this cycle are adjacent. Chessboard colours alternate along it. Two missing opposite-colour squares split the cycle into two paths, each with an even number of remaining squares. Pair consecutive squares along each path to obtain a domino tiling.

4 Induction And Monovariants

Basic – Polygon Triangulation

Base case $n = 3$: a triangle is already triangulated, with $3 - 2 = 1$ triangle. Assume any convex k -gon triangulates into $k - 2$ triangles for $3 \leq k < n$. For an n -gon, pick any diagonal that splits it into a k -gon and an $(n - k + 2)$ -gon with $3 \leq k \leq n - 1$. By induction, the triangulation has $(k - 2) + ((n - k + 2) - 2) = n - 2$ triangles: the triangles on the two sides already account for the entire polygon. All triangulations have the same count.

Key insight: Induction on the number of sides; any diagonal reduces the problem to smaller polygons.

Easy – Induction Inequality

Base case: $4! = 24 > 16 = 2^4$. Assume $n! > 2^n$ for some $n \geq 4$. Then

$$(n + 1)! = (n + 1)n! > (n + 1)2^n > 2 \cdot 2^n = 2^{n+1}.$$

Medium – Infected Chessboard

The answer is n . For sufficiency, infect the main diagonal. Induct on $|i - j|$: each off-diagonal cell has two neighbours with a smaller value of $|i - j|$, so infection spreads to every cell.

For necessity, measure the perimeter of the union of infected unit squares, including boundaries around holes. Infecting a new square with at least two infected neighbours changes this perimeter by $4 - 2k \leq 0$, where $k \geq 2$ is its number of infected neighbours. Fewer than n initial squares have perimeter at most $4(n - 1) < 4n$, whereas the fully infected board has perimeter $4n$. Thus fewer than n squares cannot suffice.

Hard – Parliament Pacification

Start with any split into two chambers. If someone has at least two enemies in their chamber, move them to the other chamber. They have at most three enemies total, so after moving they have at most one enemy in the new chamber. The total number of same-chamber enemy pairs decreases, so the process terminates. At termination, no member has more than one enemy in their chamber.

Very Hard – Pebble Spreading

The possible values are $n = 1$ and $n = 2$. One move at $(0, 0)$ clears the first diagonal. To clear the first two, move pebbles in this order: $(0, 0)$, $(1, 0)$, $(1, 1)$, $(0, 1)$. Each move has two empty destination cells, and no pebbles remain on levels $i + j < 2$.

Suppose a finite sequence clears the first three levels. Let m_{ij} be the number of moves made at (i, j) , and let f_{ij} and p_{ij} be its initial and final occupancies (each 0 or 1). Counting pebbles that arrive at and leave a cell gives

$$m_{ij} = f_{ij} + m_{i-1,j} + m_{i,j-1} - p_{ij}, \quad m_{-1,j} = m_{i,-1} = 0.$$

Here $f_{00} = 1$ and all other $f_{ij} = 0$. As $p_{ij} = 0$ on levels 0, 1, and 2, we must have

$$m_{00} = m_{10} = m_{01} = m_{20} = m_{02} = 1, \quad m_{11} = 2.$$

On every later level, $p_{ij} \leq 1$, so $m_{ij} \geq m_{i-1,j} + m_{i,j-1} - 1$. In particular, the adjacent cells $(2, 1)$ and $(1, 2)$ must each be moved at least twice. Whenever two adjacent cells on one level are each moved at least twice, their common child on the next level must be moved at least three times, while its two neighbouring cells must each be moved at least once. The following level therefore has two adjacent cells each moved at least $3 + 1 - 1 = 3$ times. Repeating this two-level argument forces moves on infinitely many levels, contradicting the assumed finite sequence. Thus even the first three levels cannot be cleared, and neither can any larger number.

